

MATH3303 – Lecture 34

Eisenstein's Criterion; Irreducibles in $\mathbb{Z}[x]$

Review Notes

Reference: Lee, Abstract Algebra, Section 11.2

Overview

This lecture:

- Completes the proof from Lecture 33 that reducibility of $f \in \mathbb{Z}[x]$ in $\mathbb{Q}[x]$ can be scaled back into a $\mathbb{Z}[x]$ factorisation.
- Proves **Eisenstein's Criterion**, a powerful irreducibility test in $\mathbb{Q}[x]$.
- Classifies the irreducibles in $\mathbb{Z}[x]$ (Lemma 11.3): they are either prime constants or primitive polynomials irreducible over $\mathbb{Q}$.

1 Proof of the $\mathbb{Z}[x]/\mathbb{Q}[x]$ Theorem

Recall (Lecture 33, Theorem 4.2): if $f(x) \in \mathbb{Z}[x]$ and $f(x) = g(x)h(x)$ with $g(x), h(x) \in \mathbb{Q}[x]$, then there exists a positive $q \in \mathbb{Q}$ with $qg(x), \frac{1}{q}h(x) \in \mathbb{Z}[x]$.

Proof. **Case 1: $f(x)$ is primitive.**

Choose positive integers $a, b \in \mathbb{Z}$ such that $ag(x), bh(x) \in \mathbb{Z}[x]$. Then

$$ab f(x) = (ag(x))(bh(x)) \in \mathbb{Z}[x].$$

Let $c := c(ag(x))$ and $d := c(bh(x))$ (the contents). Then

$$\frac{a}{c}g(x), \quad \frac{b}{d}h(x) \in \mathbb{Z}[x],$$

and both are primitive. By Gauss's Lemma (Lecture 33), their product

$$\frac{ab}{cd}g(x)h(x) = \frac{ab}{cd}f(x)$$

is primitive. Therefore $c(\frac{ab}{cd}f(x)) = 1$, i.e. $c(abf(x)) = cd$.

On the other hand, since $f(x)$ is primitive, $c(abf(x)) = ab$. Hence $ab = cd$.

Set $q := \frac{a}{c}$, so $\frac{b}{d} = \frac{1}{q}$. Then $qg(x), \frac{1}{q}h(x) \in \mathbb{Z}[x]$, as required.

Case 2: $f(x)$ is not primitive.

If $f(x) = 0$, the result is trivial.

If $f(x) \neq 0$, let $k := c(f(x))$. Then $f(x) = kf_1(x)$, where $f_1(x) \in \mathbb{Z}[x]$ is primitive. We have

$$f_1(x) = \left(\frac{1}{k}g(x)\right)h(x).$$

By the argument in Case 1 applied to the primitive polynomial f_1 , there is a positive $q \in \mathbb{Q}$ such that

$$\frac{q}{k}g(x), \quad \frac{1}{q}h(x) \in \mathbb{Z}[x].$$

But then also $qg(x) = k \cdot \frac{q}{k}g(x) \in \mathbb{Z}[x]$, since $k \in \mathbb{Z}$. □

2 Eisenstein's Criterion

Theorem 2.1: Eisenstein's Criterion (from lecture; cf. Lee, Thm. 11.5)

Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$ with $n \geq 1$ and $a_n \neq 0$. Suppose there is a prime p such that

$$p \mid a_i \text{ for } 0 \leq i < n, \quad p \nmid a_n, \quad p^2 \nmid a_0.$$

Then $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Proof. By the previous theorem (and Lemma 11.3 below), we may suppose for contradiction that $f(x) = g(x)h(x)$ with

$$g(x) = b_0 + b_1x + \cdots + b_mx^m \in \mathbb{Z}[x], \quad 1 \leq m < n,$$

$$h(x) = c_0 + c_1x + \cdots + c_kx^k \in \mathbb{Z}[x], \quad 1 \leq k < n.$$

Consider the map $\bar{\pi} : \mathbb{Z}[x] \rightarrow \mathbb{Z}_p[x]$ reducing coefficients mod p . In $\mathbb{Z}_p[x]$,

$$\bar{\pi}(f) = \overline{a_n}x^n,$$

since $p \mid a_i$ for $0 \leq i < n$. This is a (constant times a) monic irreducible-power, namely $\overline{a_n}x^n$ – the irreducibles are all equal to x .

Recall $\mathbb{Z}_p$ is a field, so $\mathbb{Z}_p[x]$ is a PID, hence a UFD.

Since $f = gh$, we have $\bar{\pi}(f) = \bar{\pi}(g)\bar{\pi}(h) = \overline{a_n}x^n$. By unique factorisation in $\mathbb{Z}_p[x]$,

$$\bar{\pi}(g) = \overline{b_m}x^m \quad \text{and} \quad \bar{\pi}(h) = \overline{c_k}x^k.$$

This means all lower coefficients of g and h are divisible by p . In particular, $p \mid b_0$ and $p \mid c_0$.

But then $a_0 = b_0c_0$ is divisible by p^2 , contradicting $p^2 \nmid a_0$. $\square$

Example 2.1: $x^n - 15$ is irreducible in $\mathbb{Q}[x]$ (from lecture; cf. Lee, Example 11.10)

For any $n \in \mathbb{N}$, consider $f(x) = x^n - 15$.

Take $p = 3$. Then $3 \mid (-15)$, $3 \nmid 1$, and $3^2 = 9 \nmid (-15)$. By Eisenstein's Criterion, $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Example 2.2: $4x - 8$ is irreducible in $\mathbb{Q}[x]$ but reducible in $\mathbb{Z}[x]$ (from lecture; cf. Lee, Example 11.11)

$4x - 8 = 4(x - 2)$, where the factor $x - 2$ has degree 1, hence is irreducible in $\mathbb{Q}[x]$. The constant 4 is a unit in $\mathbb{Q}[x]$, so the factorisation is trivial – $4x - 8$ is irreducible in $\mathbb{Q}[x]$.

But in $\mathbb{Z}[x]$, $4x - 8 = 4 \cdot (x - 2)$ where neither 4 nor $x - 2$ is a unit ($\mathbb{Z}[x]^\times = \{\pm 1\}$). Hence $4x - 8$ is reducible in $\mathbb{Z}[x]$.

3 Classification of Irreducibles in $\mathbb{Z}[x]$

Lemma 3.1: Irreducibles in $\mathbb{Z}[x]$ (from lecture; cf. Lee, Lemma 11.3)

Let $f(x) \in \mathbb{Z}[x]$. Then $f(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if one of:

- (1) $f(x) = \pm p$, where p is a prime in $\mathbb{Z}$; or
- (2) $f(x)$ is a primitive irreducible polynomial in $\mathbb{Q}[x]$.

Side notes from the lecture.

- The units in $\mathbb{Z}[x]$ are ± 1 . (These are also units in $\mathbb{Q}[x]$, and the units in $\mathbb{Q}[x]$ are exactly the nonzero constants.)
- “Invertible in $\mathbb{Z}[x]$ ” means ± 1 ; not all nonzero constants.

Proof. ($\Rightarrow$) Suppose $f(x)$ is irreducible in $\mathbb{Z}[x]$.

Case (a): $f(x) = c$ is a constant. Suppose $f(x) = c \in \mathbb{Z}$. If $c \neq \pm p$ for any prime p , then c has a nontrivial factorisation in $\mathbb{Z}$, hence in $\mathbb{Z}[x]$. This cannot happen since f is irreducible. So $c = \pm p$ for some prime p .

Case (b): $\deg(f(x)) \geq 1$. If $c(f) > 1$, then

$$f(x) = c(f) \cdot \left(\frac{1}{c(f)} f(x) \right)$$

is a nontrivial factorisation in $\mathbb{Z}[x]$ (both factors nonunits), contradicting irreducibility. So $c(f) = 1$, i.e. $f(x)$ is primitive.

If $f(x)$ were reducible in $\mathbb{Q}[x]$: write $f(x) = g(x)h(x)$ with $g(x), h(x) \in \mathbb{Q}[x]$ nonconstants. By Lecture 33’s theorem (proved above), there exists a positive $q \in \mathbb{Q}$ such that

$$qg(x), \frac{1}{q}h(x) \in \mathbb{Z}[x].$$

Then $f(x) = (qg(x))(\frac{1}{q}h(x))$ is a factorisation in $\mathbb{Z}[x]$ with both factors of positive degree (hence nonunits). So $f(x)$ is reducible in $\mathbb{Z}[x]$. Contradiction.

Hence $f(x)$ is a primitive irreducible polynomial in $\mathbb{Q}[x]$. □

Note. The ($\Leftarrow$) direction (which we have not written out in the lecture’s proof above) is straightforward: a prime $\pm p$ has no nontrivial integer factorisation; and a primitive polynomial irreducible in $\mathbb{Q}[x]$ cannot factor nontrivially in $\mathbb{Z}[x]$ either, since any constant factor must equal ± 1 by primitivity, and any positive-degree factorisation would persist into $\mathbb{Q}[x]$.

Checklist of Skills

- ☐ Prove the scaling theorem: a $\mathbb{Q}[x]$ -factorisation of $f \in \mathbb{Z}[x]$ scales to a $\mathbb{Z}[x]$ -factorisation (primitive case via Gauss's Lemma, then reduce to the primitive case).
- ☐ State and prove Eisenstein's Criterion via reduction mod p and unique factorisation in $\mathbb{Z}_p[x]$.
- ☐ Apply Eisenstein's Criterion to show $x^n - p$ is irreducible in $\mathbb{Q}[x]$ for any prime p .
- ☐ Explain why $4x - 8$ is irreducible in $\mathbb{Q}[x]$ but reducible in $\mathbb{Z}[x]$ (units differ in the two rings).
- ☐ State the lemma classifying irreducibles in $\mathbb{Z}[x]$: either prime constants $\pm p$, or primitive polynomials irreducible in $\mathbb{Q}[x]$.
- ☐ Prove the classification by handling the constant case, the content reduction, and the lift from $\mathbb{Q}[x]$ -reducibility.
- ☐ Identify the units in $\mathbb{Z}[x]$ as ± 1 , and in $\mathbb{Q}[x]$ as the nonzero constants $\mathbb{Q}^\times$.